# Trustworthy Deep Learning for Chest X-ray Disease Detection

## 1. Problem Motivation

Chest X-ray (CXR) is the most common medical imaging examination worldwide, and deep learning models can classify chest diseases with reported accuracies above 95%. However, recent research shows that many of these results are misleading. Models frequently exploit hidden dataset artefacts such as hospital source markers or acquisition-device signatures rather than genuine lung pathology. This is called shortcut learning. When tested on data from unseen hospitals, performance drops dramatically, with AUC falling from above 0.99 to around 0.76.

A CXR model intended for clinical use must satisfy three conditions beyond accuracy:

(1) It must focus on the diseased lung region rather than image artefacts.

(2) Its confidence scores must be calibrated so clinicians can trust predicted probabilities.

(3) It must generalise to unseen hospital data.

Existing studies address these properties separately. No work combines accuracy, calibration, explanation faithfulness, computational efficiency, and cross-dataset robustness into one controlled comparison, nor proposes a lightweight architectural fix for shortcut learning that generalises across model families. This project addresses both gaps.

## 2. Identified Datasets

The primary dataset is the COVID-19 Radiography Database (Kaggle), containing approximately 21,000 images across four classes: COVID-19, Normal, Viral Pneumonia, and Lung Opacity. It is publicly available, well-structured, and known to contain source bias across classes, making it suitable for studying shortcut learning. The external dataset is the RSNA Pneumonia Detection dataset (Kaggle), with frontal chest X-rays labelled Pneumonia and Normal from different hospital sources. This is used exclusively for out-of-distribution robustness evaluation. Models are never trained on it, and the accuracy drop between internal and external results serves as the robustness measure.

## 3. Baseline Model

Since the data type is images, the baseline architecture is a Convolutional Neural Network (CNN) following CS3631 baseline requirements. We select DenseNet121 as the primary baseline due to its established performance in medical imaging through dense feature reuse, with approximately 8.0M parameters. The model uses ImageNet-pretrained weights as initialisation and replaces the final classification layer with a 4-class output head.

Images are resized to 224x224, converted to 3-channel, and normalised with ImageNet statistics. Data is split 70/15/15 (train/validation/test) using stratified sampling with a fixed seed. Training augmentation includes random rotation, horizontal flip, brightness/contrast jitter, and random cropping. The model is implemented in PyTorch using the timm library with AdamW optimiser, cross-entropy loss, and class weighting. All experiments run on Google Colab and Kaggle free GPUs.

Initial experiments with default hyperparameters on pre-trained models yield limited classification accuracy on chest X-ray data. To address this, we conduct systematic hyperparameter tuning covering learning rate, batch size, optimiser choice (Adam, AdamW, SGD with momentum), weight decay, and learning rate schedulers (cosine annealing, step decay, reduce-on-plateau).

We employ a two-phase transfer learning strategy:

Phase 1 freezes the backbone and trains only the classification head.

Phase 2 unfreezes the final dense blocks for end-to-end fine-tuning at a reduced learning rate.

Early stopping on validation loss prevents overfitting. This tuning is applied identically across all models for fair comparison.

## 4. Proposed Contribution

Our contribution is twofold: a novel lightweight architectural addition to suppress shortcut learning, and a unified multi-axis trustworthiness benchmark to evaluate it.

***4.1 Shortcut-Suppression Module.*** We propose a lightweight, model-agnostic module that can be inserted into any backbone to force the network to attend to clinically relevant lung regions rather than dataset artefacts. We investigate three candidate designs:

(a) a Lung-Region Attention Module - A spatial attention block guided by an automatically generated lung mask that re-weights feature maps to suppress background and artefact regions.

(b) an Auxiliary Segmentation Head - A secondary decoder branch attached to intermediate features that jointly predicts a lung segmentation mask during training, acting as an anatomical regulariser that steers the shared backbone toward lung-relevant features.

(c) a Shortcut-Suppression Loss Term - A custom regularisation loss that penalises the model when Grad-CAM activations fall outside the lung region, directly optimising for explanation faithfulness during training.

We first develop and remove each candidate against the vanilla DenseNet121 baseline to isolate its effect. The winning design is then plugged into all four architectures (ResNet50, DenseNet121, EfficientNet-B0, ViT-Base) to demonstrate that the fix generalises across CNN and Transformer families, not just one model.

***4.2 Five-Axis Trustworthiness Benchmark.*** The multi-architecture comparison serves as the evaluation framework for our proposed module. We benchmark all models, with and without the shortcut-suppression module across five axes, They are,

- Accuracy.
- confidence calibration.
- explanation faithfulness.
- computational efficiency.
- cross-dataset robustness.

This controlled before-and-after comparison across architectures produces strong evidence of whether the proposed module genuinely reduces shortcut reliance or merely shifts the problem.

## 5. Experimental Plan

The experiments are structured in three stages.

- Stage 1 (Baseline Establishment):- train all four vanilla architectures under an identical pipeline (same preprocessing, augmentation, optimiser, split, seed) and report baseline metrics.
- Stage 2 (Module Development and Ablation):- develop the three candidate shortcut-suppression modules on DenseNet121, ablate each component, and select the best-performing design based on improvement in explanation faithfulness and robustness without significant accuracy loss.
- Stage 3 (Cross-Architecture Generalisation):- apply the selected module to all four architectures and compare with-module versus without-module performance across all five evaluation axes.
- Classification is evaluated using accuracy, precision, recall, F1-score, ROC-AUC (per-class and macro), and confusion matrices.

- Calibration uses ECE, Brier score, and reliability diagrams before and after temperature scaling. Explainability is assessed via heatmap comparison and a quantitative lung-localisation faithfulness score.
- Robustness is measured as internal-to-external accuracy and AUC drop.
- Efficiency metrics include parameter count, FLOPs, model size, and inference time on CPU and GPU.
- DenseNet121 serves as the primary baseline. All four architectures are trained under an identical pipeline (same preprocessing, augmentation, optimiser, split, seed) so differences are attributable to architecture alone.
- Frozen versus fine-tuned transfer learning is compared for each model.
- All experiments run on freely available resources (Google Colab, Kaggle). Input resolution is 224x224 with mixed-precision training.
- No multi-GPU clusters or HPC infrastructure are required.